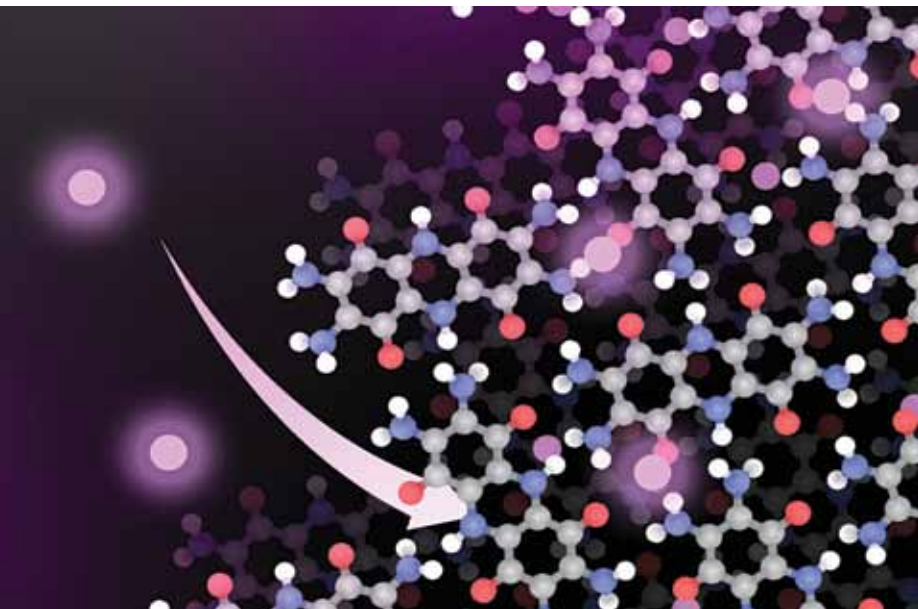


Sustainable organic cathode batteries for electric cars innovated



*A new MIT battery material could offer a more sustainable way to power electric cars
(Source: Courtesy of the researchers; Edited by MIT News)*

MIT researchers have innovated a sustainable electric vehicle battery, side-stepping the cobalt reliance plaguing many EV batteries. Their breakthrough utilises an organic cathode, composed of TAQ layers—a molecule boasting three fused hexagonal rings, resembling a graphite-like structure. TAQ's quinones act as electron reservoirs, while amines form robust hydrogen bonds, ensuring high stability and insolubility. Tests reveal conductivity and storage capacities equivalent to cobalt batteries, with faster charging rates. Incorporating cellulose and rubber fillers enhances stability and lifespan, and estimated material costs are significantly lower. Future efforts may explore replacing lithium with more abundant sodium or magnesium in these environmentally friendly batteries.

RISC-V based 'Big Chip' to explore scalability of processor performance

Researchers from the China Academy of Sciences have introduced a report on the 'Big Chip,' a chipset-based architecture aimed at exploring the scalability of processor performance. Their creation, the Zheijiang Big Chip, incorporates a RISC-V processor system with 16 chiplets containing a total of 256 cores, fabricated using a 22nm CMOS process. This system is distinguished by its network-on-chip (NoC) that facilitates symmetrical communication across 16 RISC-V processors within each chiplet, with plans to expand the architecture to support up to 100 chiplets, as detailed in the Elsevier journal *Fundamental Research*. Utilising a die-to-die interface that employs time-multiplexing for chiplet connections, the design supports a unified memory system allowing seamless memory access across any core. This time-multiplexing technique not only minimises the space required for I/O bumps but also streamlines the wiring on the interposer.



*The Big Chip is a RISC-V processor system with 256 cores
(Credit: www.electronicsforu.com)*